YEAR 10 MATHS · CHAPTER 10 · MIND MAP

Polynomials, functions and graphs

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1 Functions and notation

A function gives one y for each x. $f(a)$: substitute $x = a$.

$f(x) = 2x^2 - 3$: $f(2) = 5$, $f(-1) = -1$

Vertical line test: a function's graph is hit at most once by any vertical line.

2 Introducing polynomials

The degree is the highest power. The leading coefficient is the number in front of it.

$P(x) = 3x^4 - 2x + 7$: degree 4, leading coefficient 3, constant 7

3 Expanding and simplifying

Multiply every term by every term, then collect like terms.

$(x^2 + 2)(x - 3) = x^3 - 3x^2 + 2x - 6$

$(x + 1)^3 = x^3 + 3x^2 + 3x + 1$

4 Dividing polynomials

Long division: divide the leading terms, multiply, subtract, repeat.

$(x^3 + 2x^2 - 5x - 6) \div (x - 2) = x^2 + 4x + 3$, remainder 0

5 Remainder and factor theorems

Dividing $P(x)$ by $(x - a)$ leaves $P(a)$. If $P(a) = 0$, $(x - a)$ is a factor.

$P(x) = x^3 - 4x^2 + x + 6$: $P(2) = 0$, so $(x - 2)$ is a factor

$P(1) = 4$: dividing by $(x - 1)$ leaves 4

6 Solving polynomial equations

Factorise fully, then use the Null Factor Law.

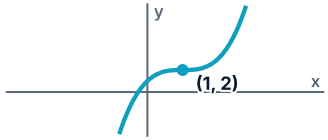
$x^3 - 4x^2 + x + 6 = (x - 2)(x - 3)(x + 1) = 0$

$x = -1, 2$ or 3

7 Cubics $y = a(x - h)^3 + k$

The point of inflection is (h, k). A negative a flips it.

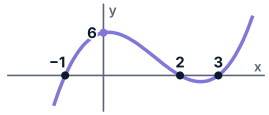
$y = (x - 1)^3 + 2$: point of inflection $(1, 2)$



8 Graphs of polynomials

Factors give the x-intercepts. A squared factor touches the axis; a cubed one flattens through it.

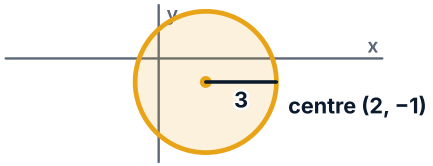
$y = (x + 1)(x - 2)(x - 3)$: x-intercepts −1, 2, 3, y-intercept 6



9 Graphs of circles

$(x - h)^2 + (y - k)^2 = r^2$: centre (h, k), radius r.

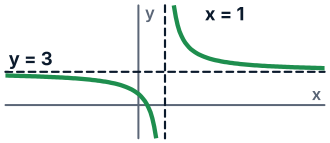
$(x - 2)^2 + (y + 1)^2 = 9$: centre $(2, -1)$, radius 3



10 Hyperbolas

$y = \frac{a}{x - h} + k$: asymptotes $x = h$ and $y = k$.

$y = \frac{2}{x - 1} + 3$: asymptotes $x = 1$ and $y = 3$



11 Further transformations

$f(x) + k$: up k. $f(x - h)$: right h. $-f(x)$: flip in the x-axis. $f(-x)$: flip in the y-axis.

$y = af(x)$: stretched from the x-axis by a factor of a